Introduction

ACOL2015: Digital Logic And System Design

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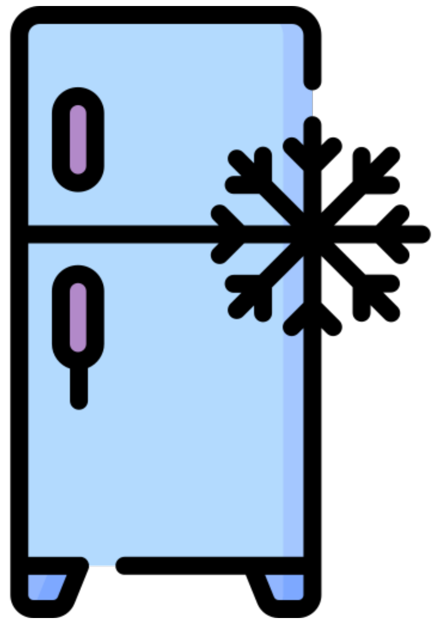
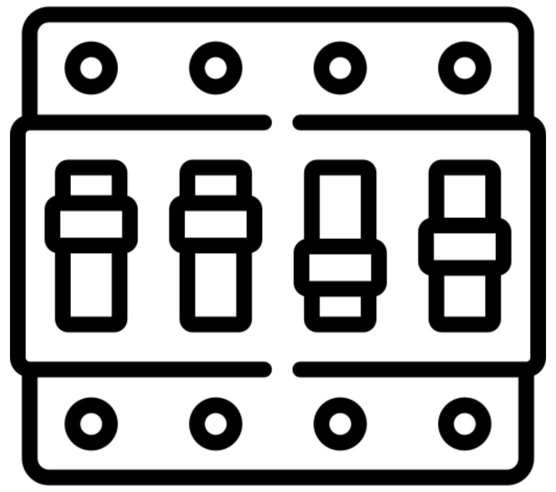
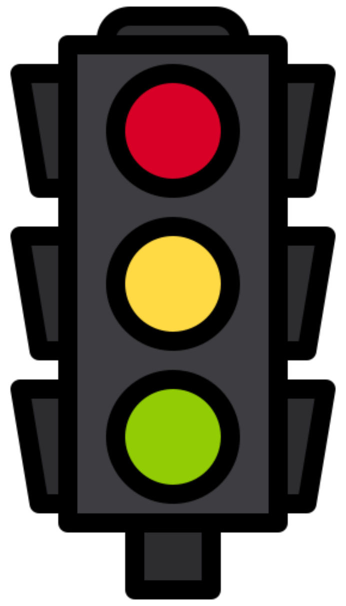
Administrivia

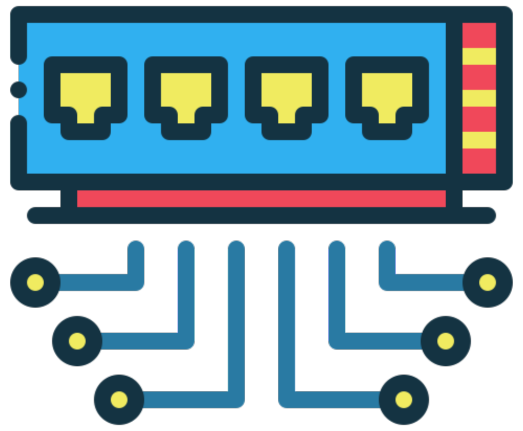
- http://abhilash-jindal.com/teach.html
- Grading criteria, TAs, late policy, audit criteria, quizzes, labs, project, discussions link

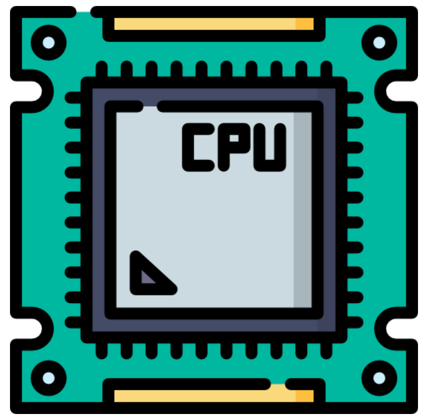
Digital systems are all around us

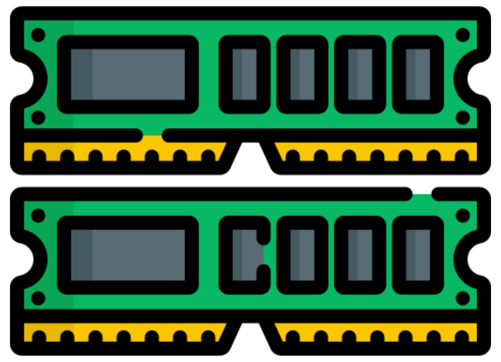


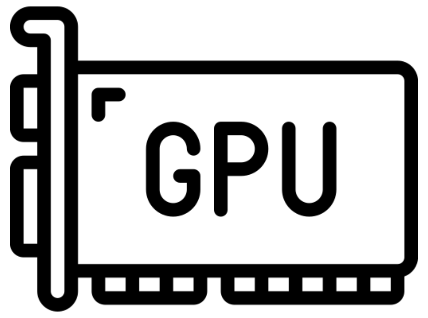










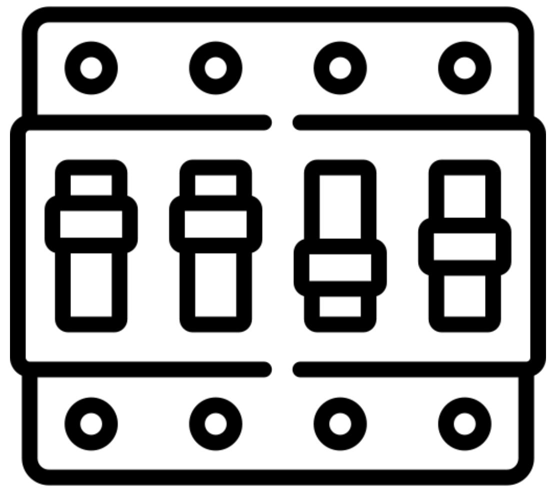


Computer

Special-purpose

General

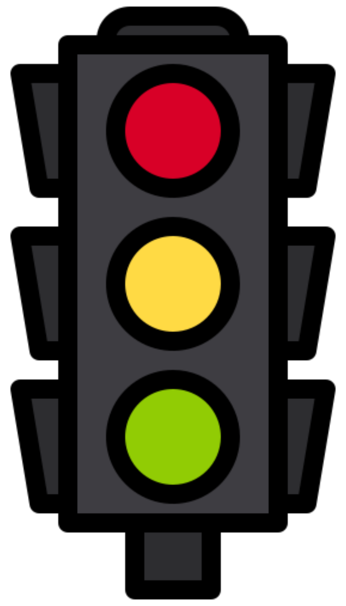
Digital systems contain circuits to implement some logic



Example: switches

- Turn the light on when any of the four switches are on

Digital systems contain circuits to implement some logic



Example: traffic lights

- Loop
  - Wait 1 minute, change light to yellow
  - Wait 10 seconds, change light to green
  - Wait 20 seconds, change light to red

Digital systems contain circuits to implement some logic

Example: washing machine

- ?

Types of circuits

- Combinational Circuit
    - Outputs are a function of only the inputs. They have no *memory*
    - Example: Switches
    - Main building block: Gates

- Sequential Circuit
    - Outputs are a function of inputs and current *state*
    - Example: Traffic lights, Washing machines, CPU
    - Main building block: Gates and Flip flops

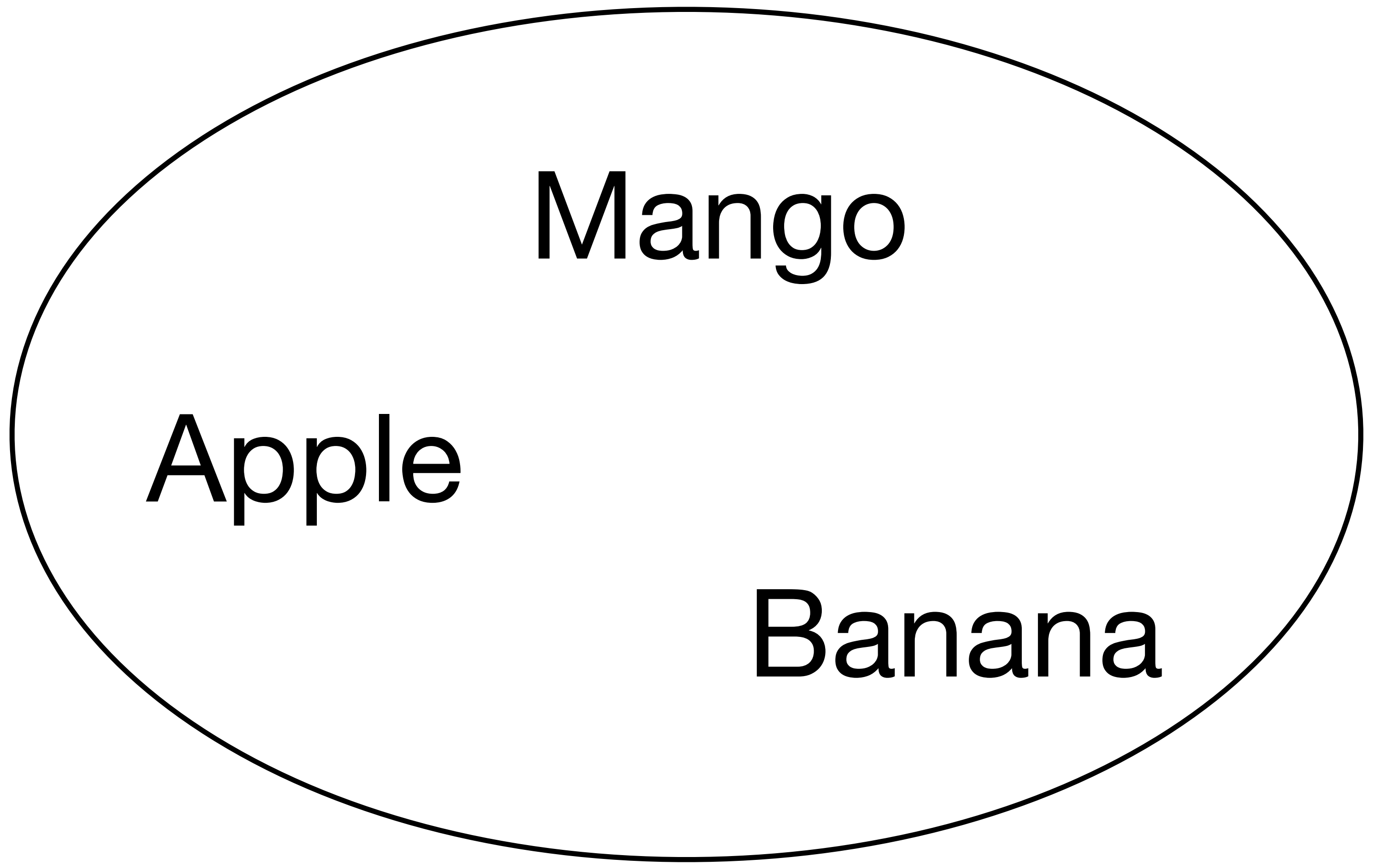
Goals

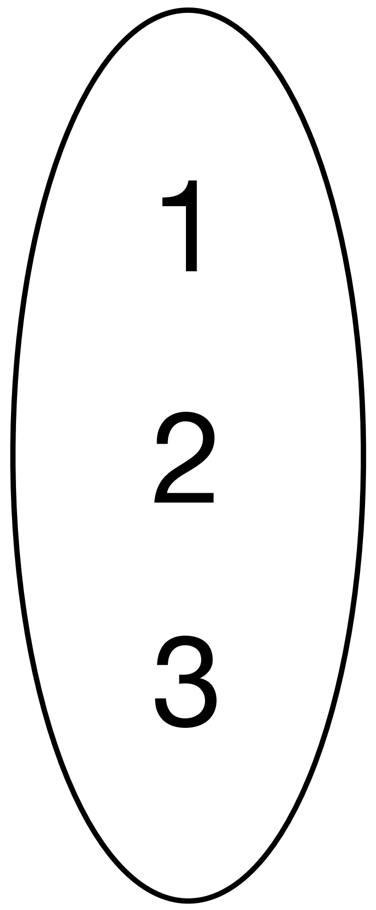
- Build basics of digital logic: Binary number system (Ch 1), Boolean algebra and logic gates (Ch 2), Gate-level minimisation (Ch 3)
- Understand foundations of circuit design: combinational logic (Ch 4), Synchronous sequential logic (Ch 5)
- Use foundations to build circuits: Registers and counters (Ch 6), Memory and programmable logic (Ch 7)
- Learn Verilog (Hardware Description Language) to build circuits

Math Basics

Set

- Collection of distinct elements
- $1 \in \{1,2,3\}$ and $4 \notin \{1,2,3\}$
- $\{1,2\} \cap \{2,3\} = ?$
- $\{1,2\} \cup \{2,3\} = ?$
- $\{1,2\} \setminus \{2,3\} = ?$
- $\mathscr{P}(\{1,2,3\})$ = ?

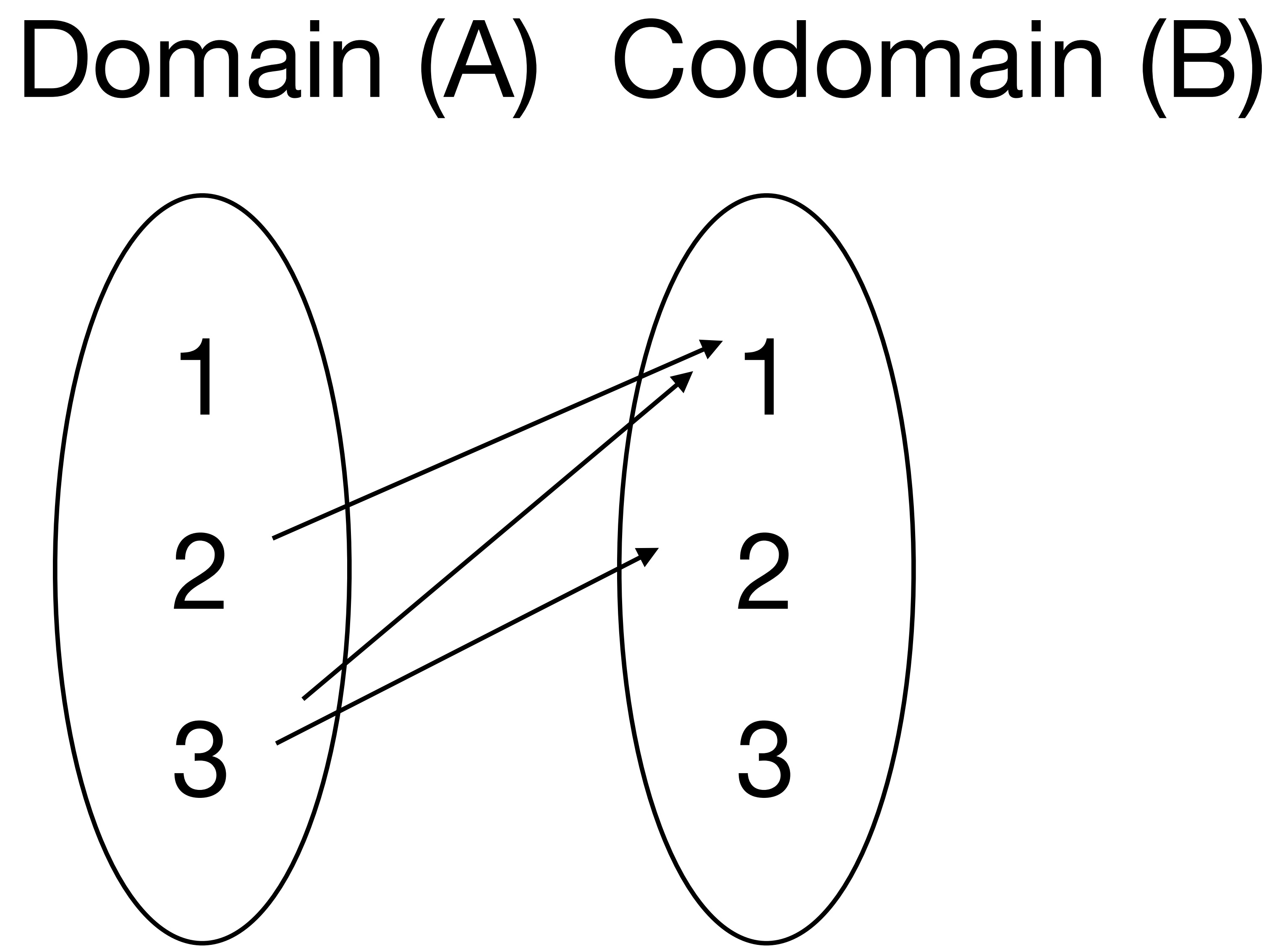




S = {1, 2, 3}

Relation

a R b



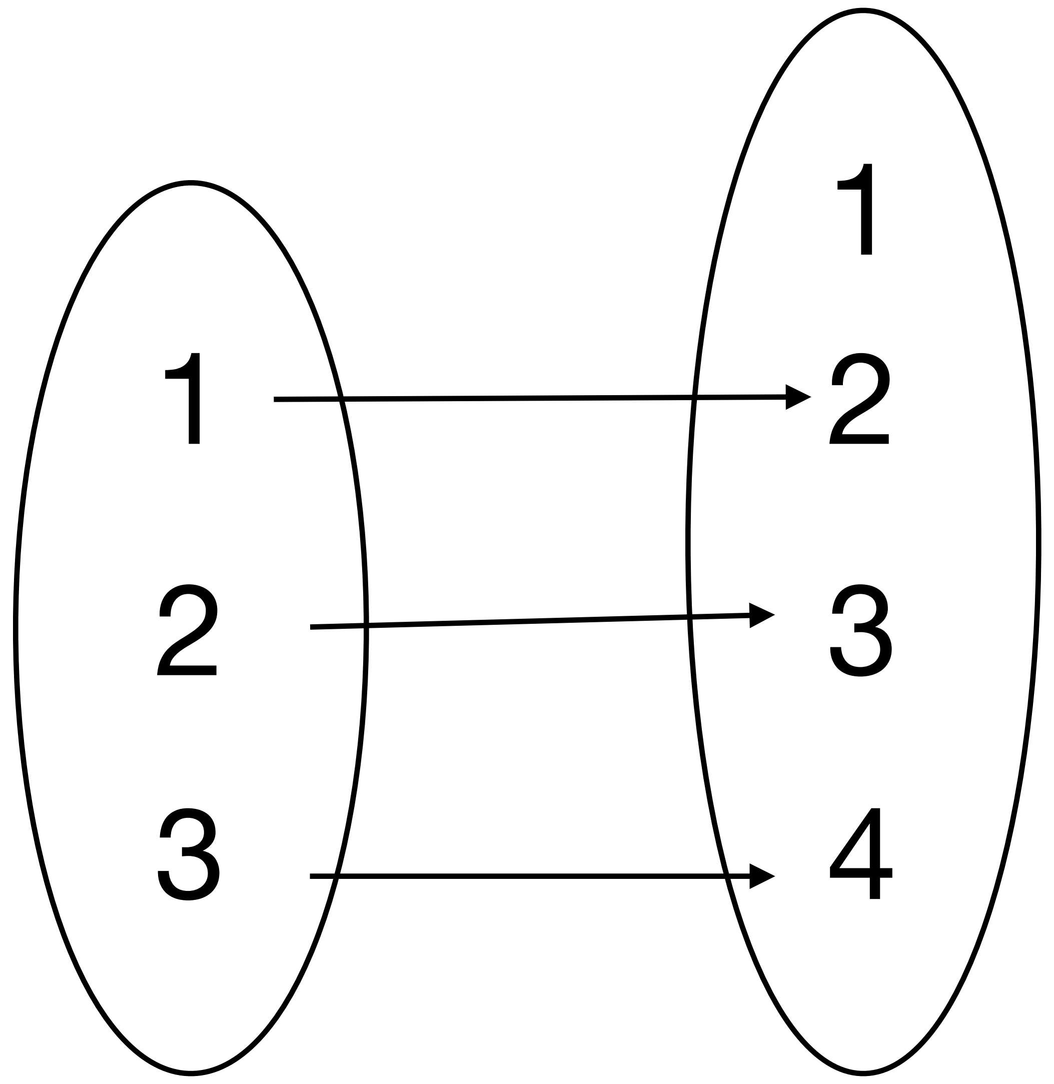
a > b = {(3, 1); (3, 2); (2, 1)}

- Reflexive relation: (a R a)
- Symmetric relation: If (a R b) then (b R a)
- Transitive relation: If (a R b) and (b R c) then (a R c)
- Equivalence relation is reflexive, symmetric, transitive
- Analyse: (a > b), (a >= b), (a = b), (b = 3), Cartesian product: A x A

Function

b = F(a)

Domain (A) Codomain (B)



b = a + 1 := {(1, 2); (2, 3); (3, 4)}

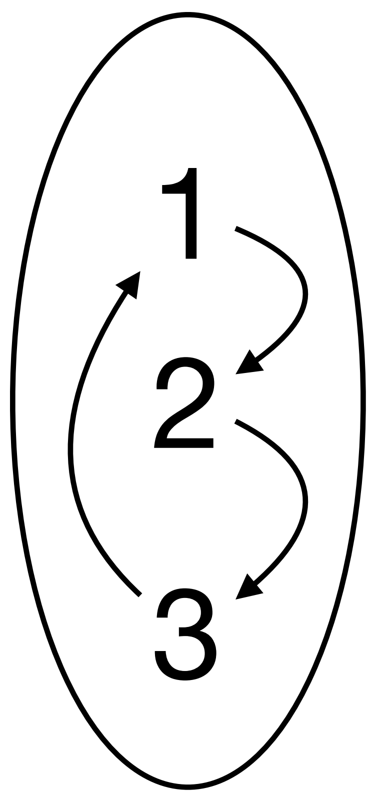
- Function is a relation where *each* element in the Domain maps to *exactly one* element in Codomain

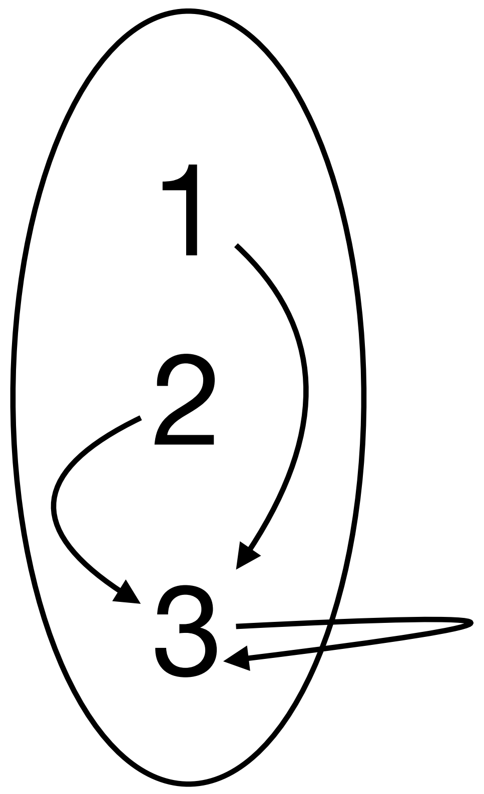
- For the given Domain, Codomain:

  - Are following relations functions?
    a > b, a = b, b = 3

  - How many functions are possible?

Unary operators





- When domain and codomain are identical, a function is also called a *unary operator.*
  - *b = (a + 1)%3; b = 3; …*
- How many unary operators are possible for the given set?

Binary operators

a op b = c. $a, b, c \in S$

- Say $a, b \in \mathbb{N}$ (natural numbers). Which of the following are binary operators?
  - Addition +
  - Subtraction -
  - Multiplication *
  - Division /
- What if $a, b \in \mathbb{R}$ (real numbers) or $\mathbb{Q}$ (rational numbers) or $\mathbb{Z}$ (integers)?

Binary Operator Properties

- Commutative: $\forall x, y \in S : x \square y = y \square x$
- Associative: $\forall x, y \in S : (x \square y) \square z = x \square (y \square z)$
- Identity element: $\exists e \in S : \forall x \in S : e \square x = x \square e = x$
- Inverses: $\forall x \in S : \exists y \in S : x \square y = e$
- Distributive: $\forall x, y, z \in S : x \square (y \Diamond z) = (x \square y) \Diamond (x \square z)$

Binary Operator Properties

- Commutative
- Associative
- Identity element
- Inverses
- Distributive

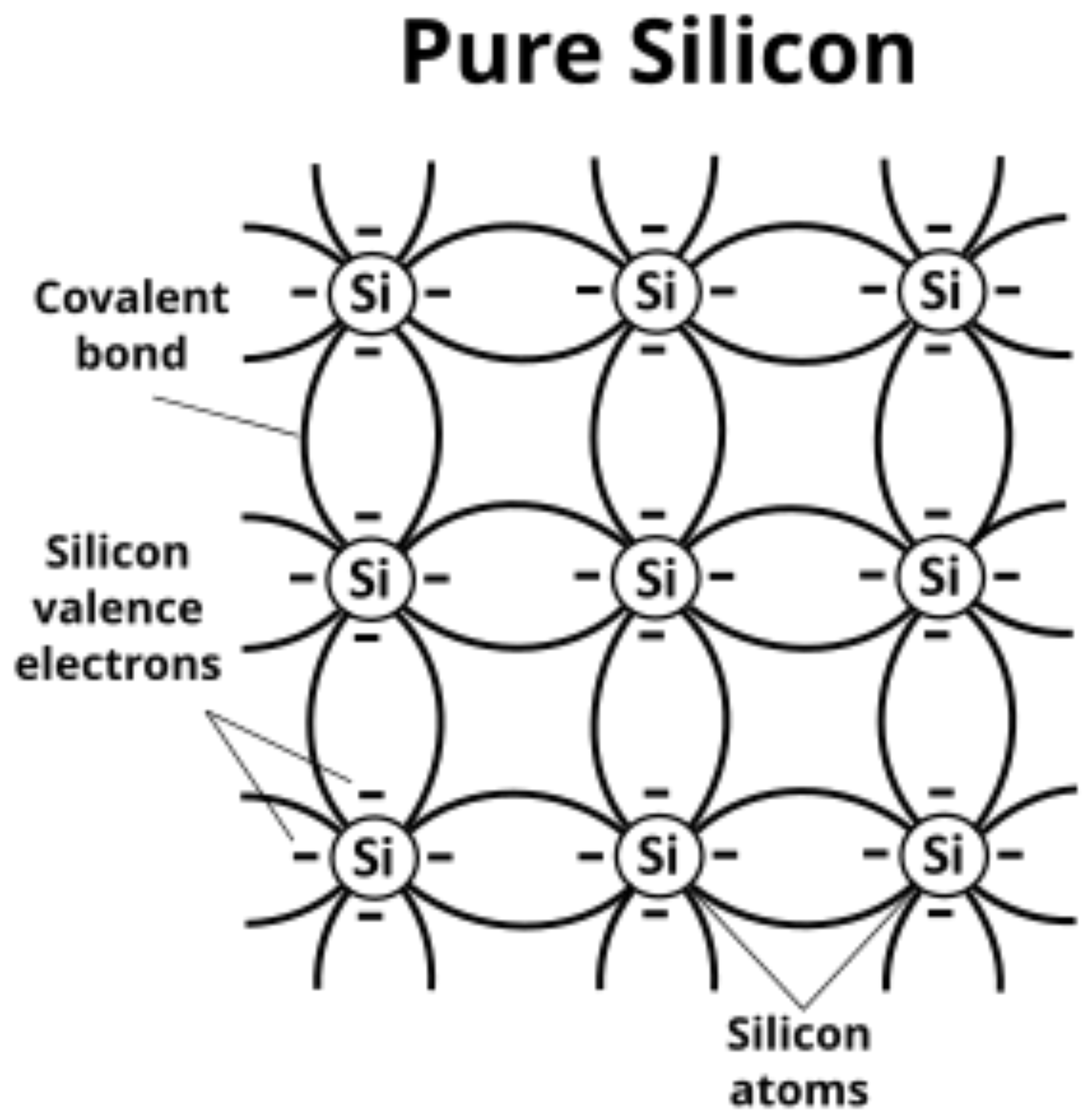
- What about +, -, *, / on real numbers?
- What about set operators $\cap$, $\cup$, $\setminus$ on a power set?

Semiconductor basics

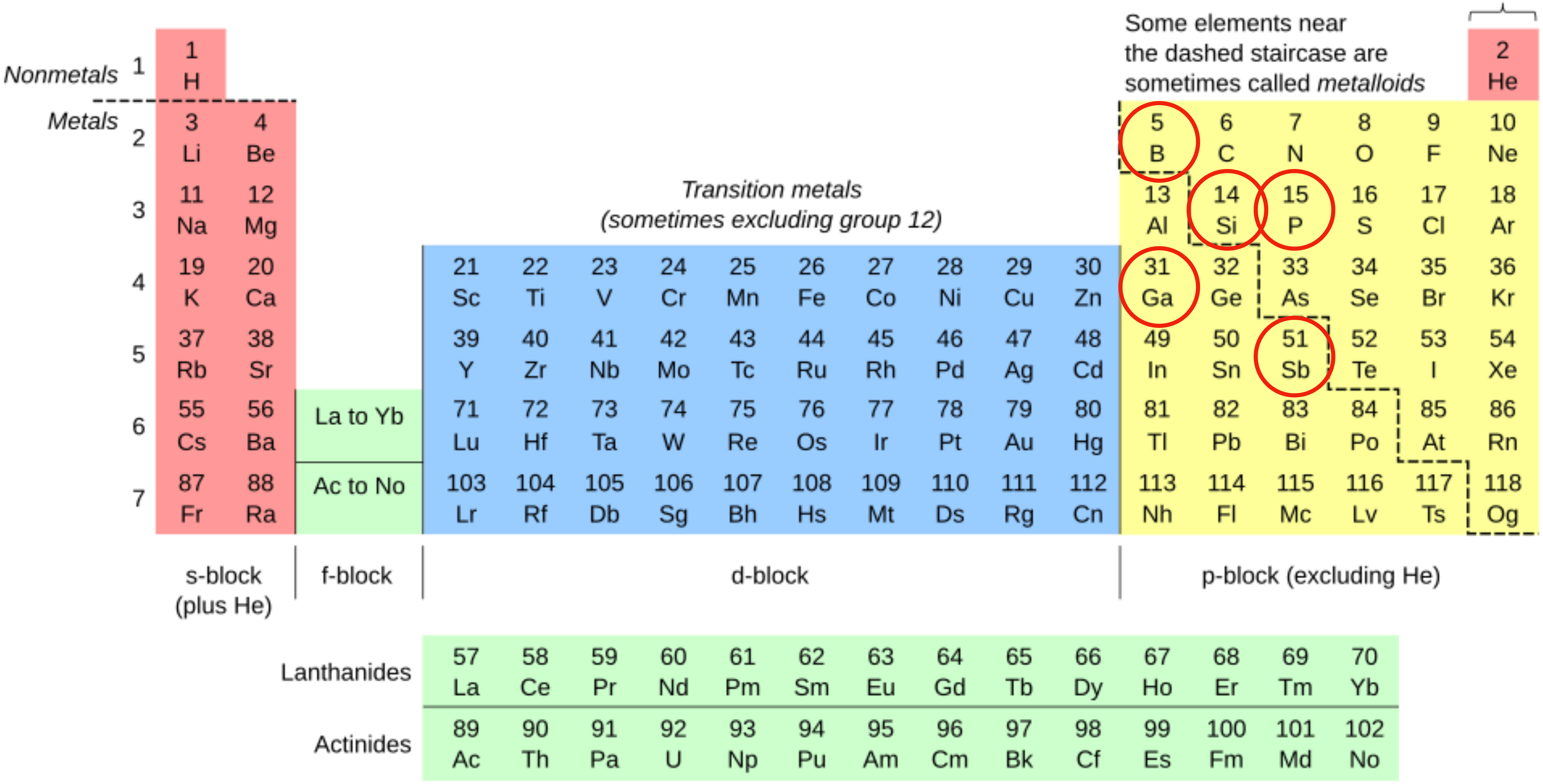
Appendix A

Silicon crystal

- Every silicon atom shares its four valence electrons with its four nearest neighbours
- No free electrons to move with an electric field

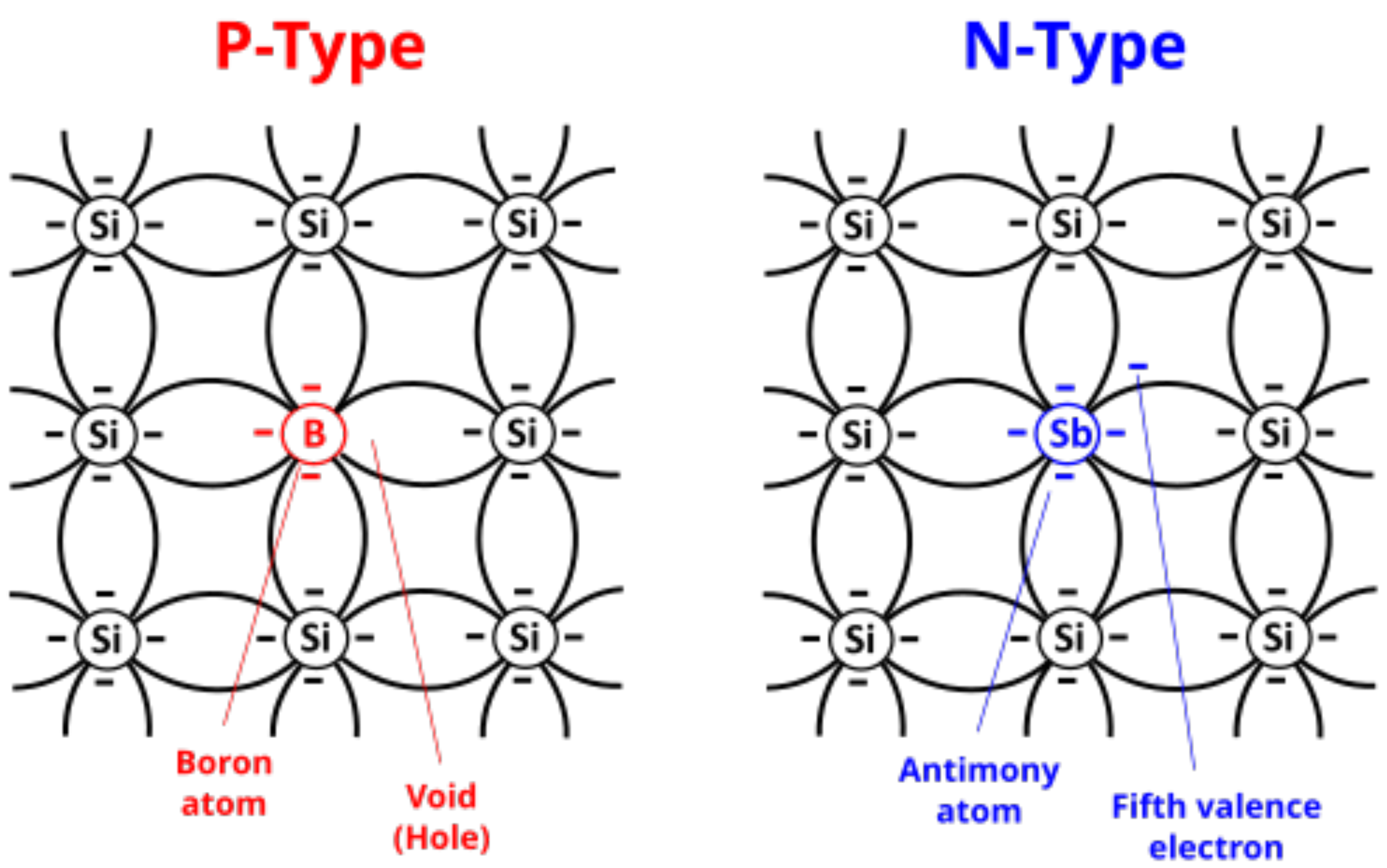


Periodic table of elements



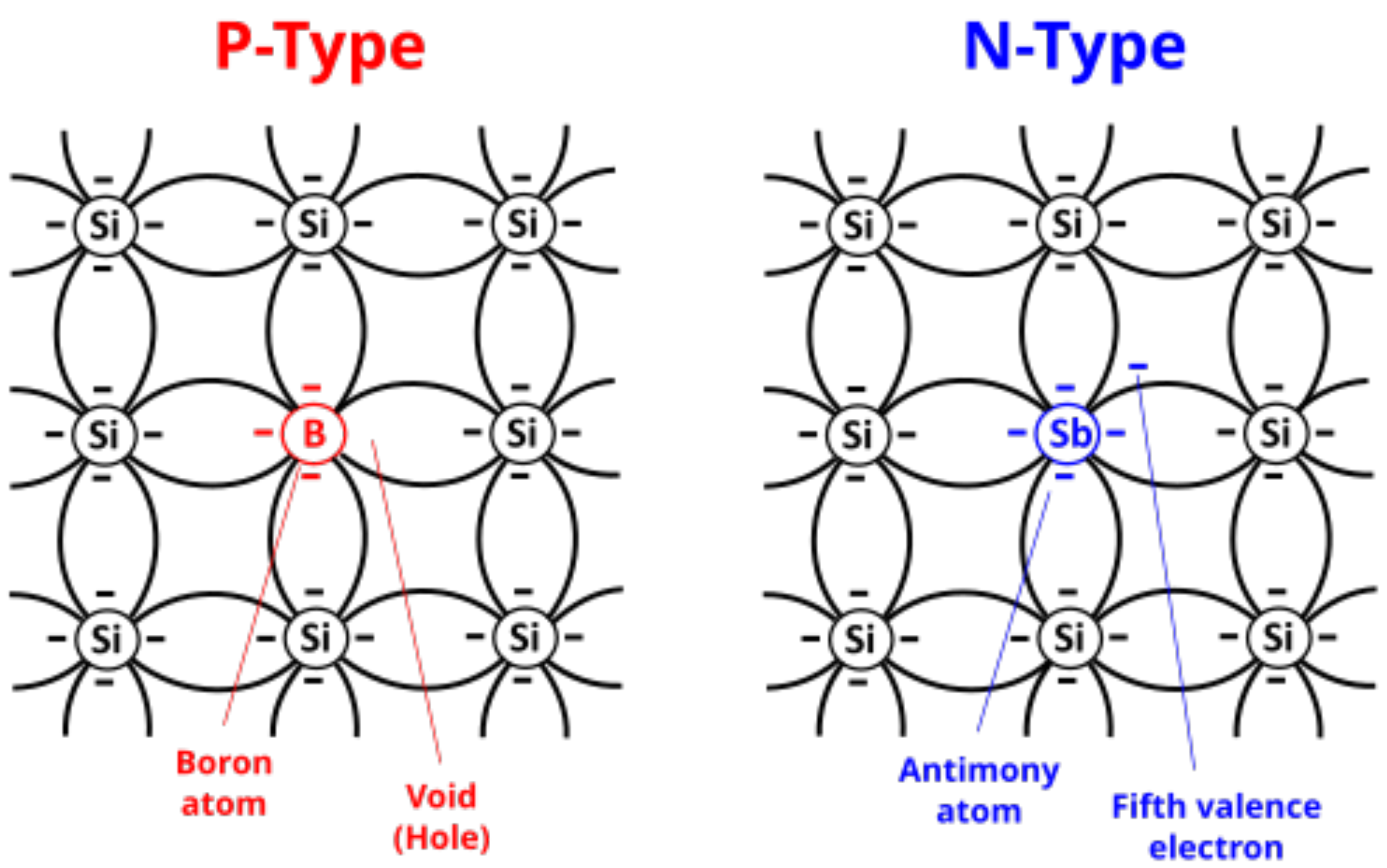
Doping

- Dopants fit easily into the silicon crystal
- P-type dopants (Boron/Gallium) have one less electron (valency of 3)
- N-type dopants (Antimony/ Phosphorous) have an extra electron (valency of 5)



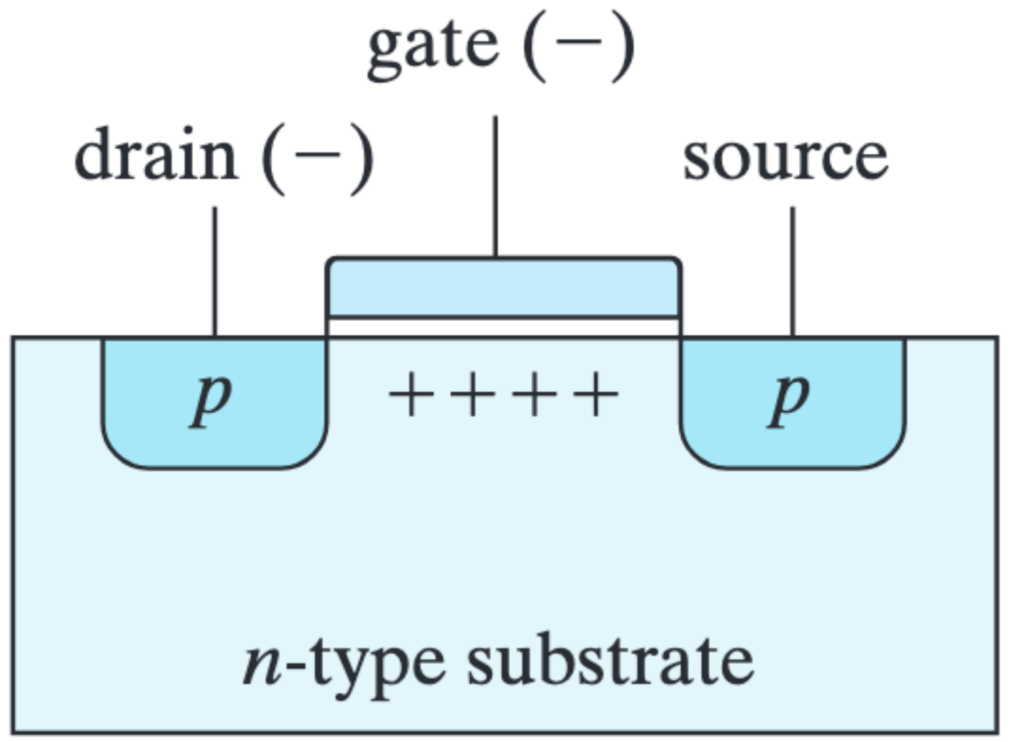
Extra electrons/holes can flow current

- Current moves in the opposite direction of electrons
- Current moves in the same direction as holes
- Metal-Oxide Semiconductor (MOS) either flow holes or electrons

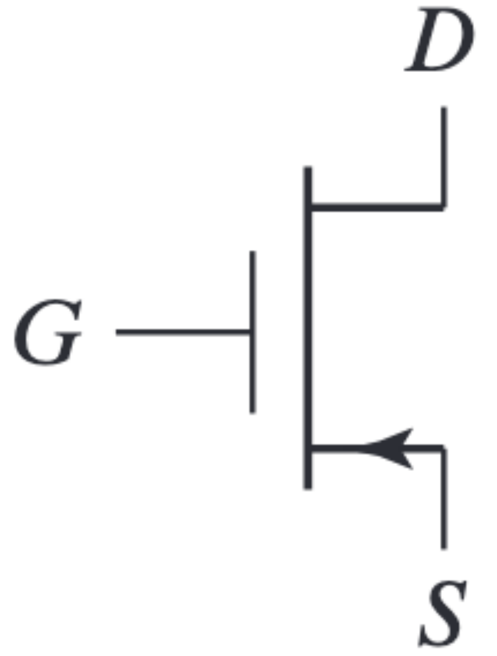


p-channel transistor

- In p-channel transistor, when gate-to-source voltage > threshold (-2V), no current flows.
    - Holes flow from source to drain
    - Current flows from source to drain

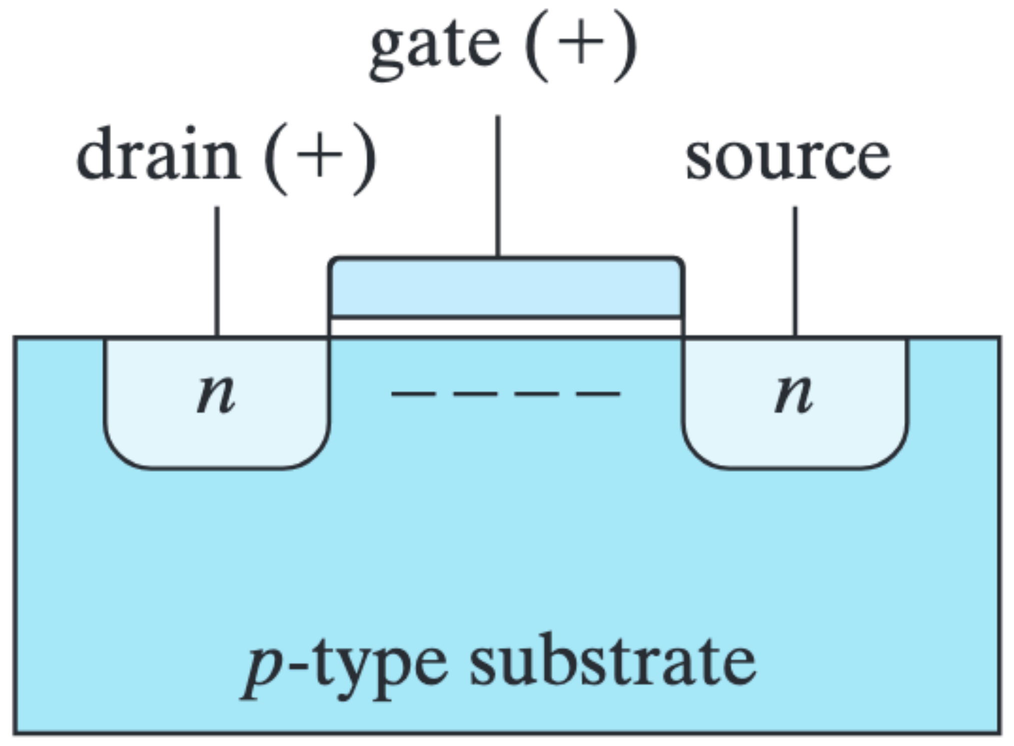


(a) p-channel

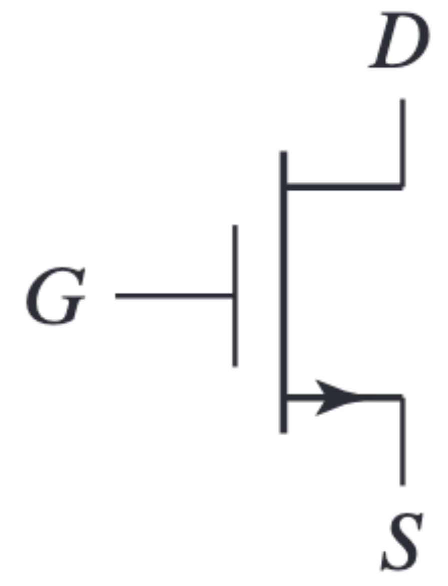


n-channel transistor

- In n-channel transistor, when gate-to-source voltage < threshold (2V), no current flows.
  - Electrons flow from source to drain
  - Current flows from drain to source

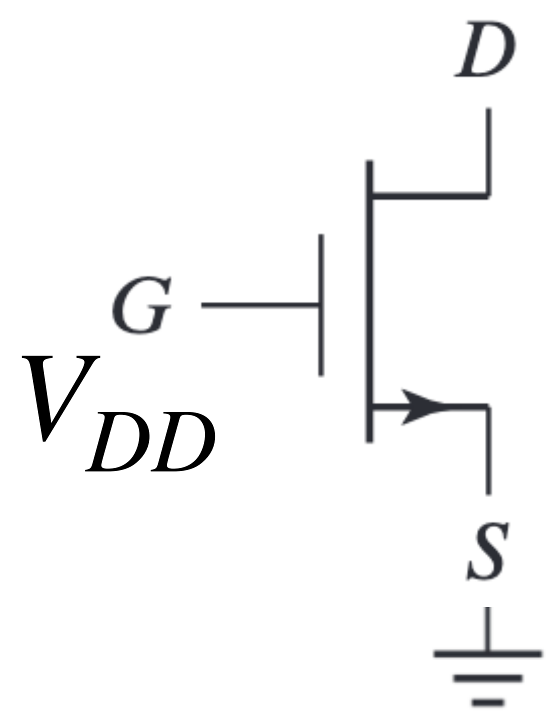


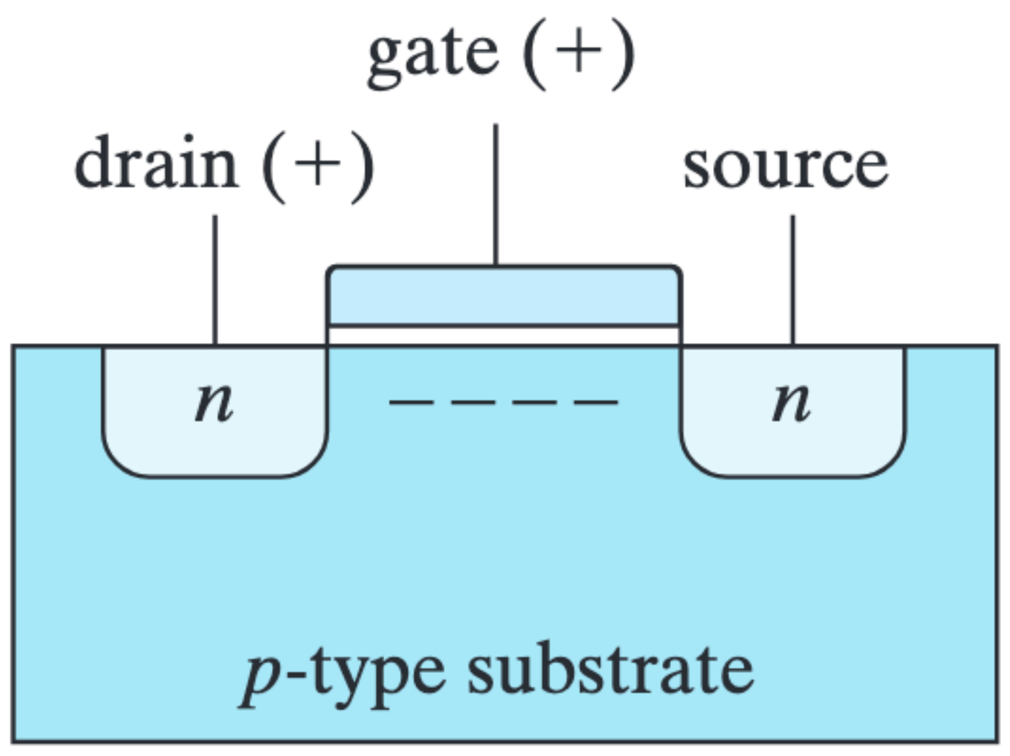
(b) *n*-channel



MOS device as a resistor

- A resistor is obtained from the MOS by *permanently biasing* the gate terminal for conduction
- In n-channel device, when gate-to-source voltage < threshold (2V), no current flows
- In n-channel resistor, gate-to-source voltage is always > threshold
- Different resistor value can be constructed during manufacturing by fixing the channel length and width of the MOS device

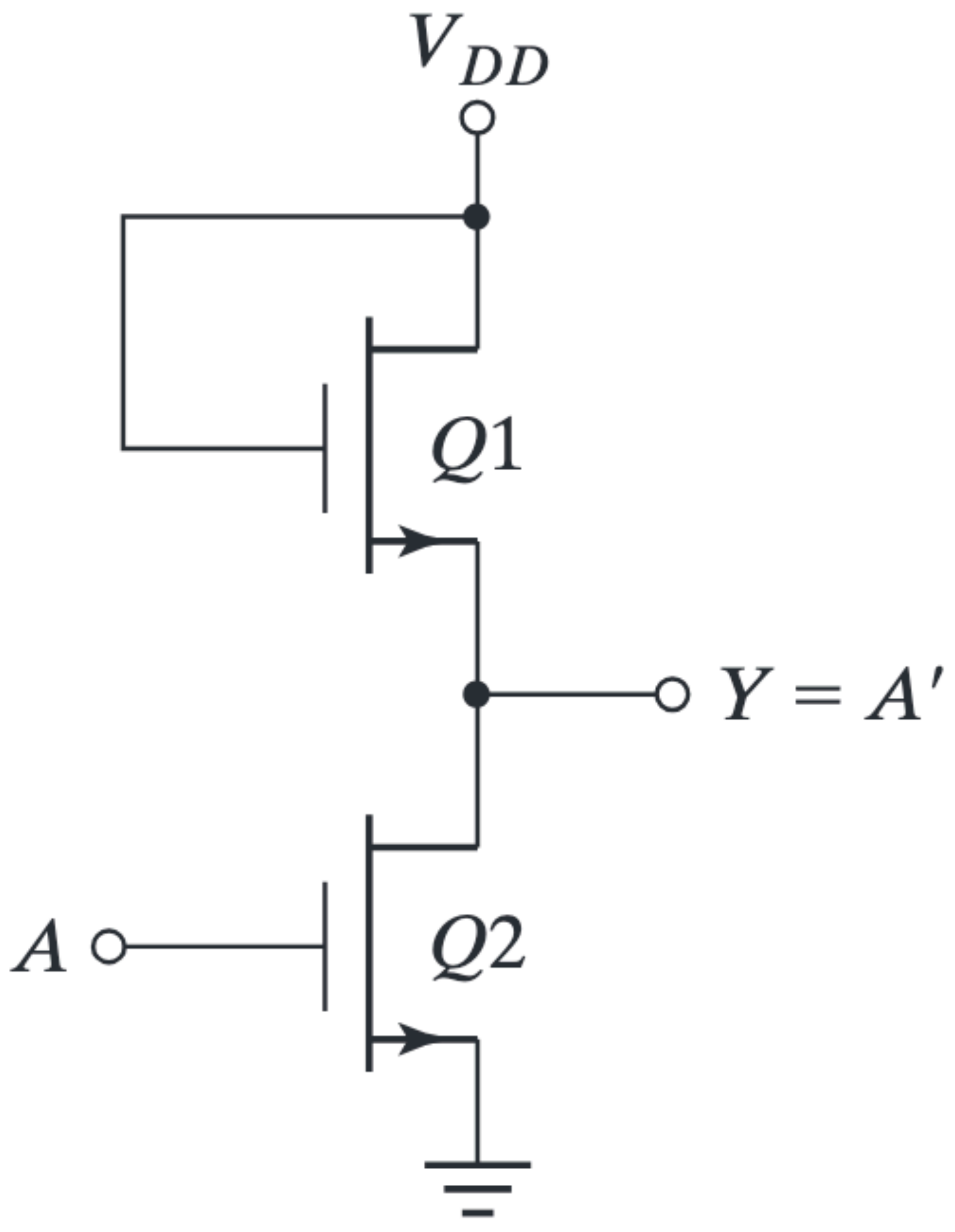




(b) *n*-channel

n-channel Inverter

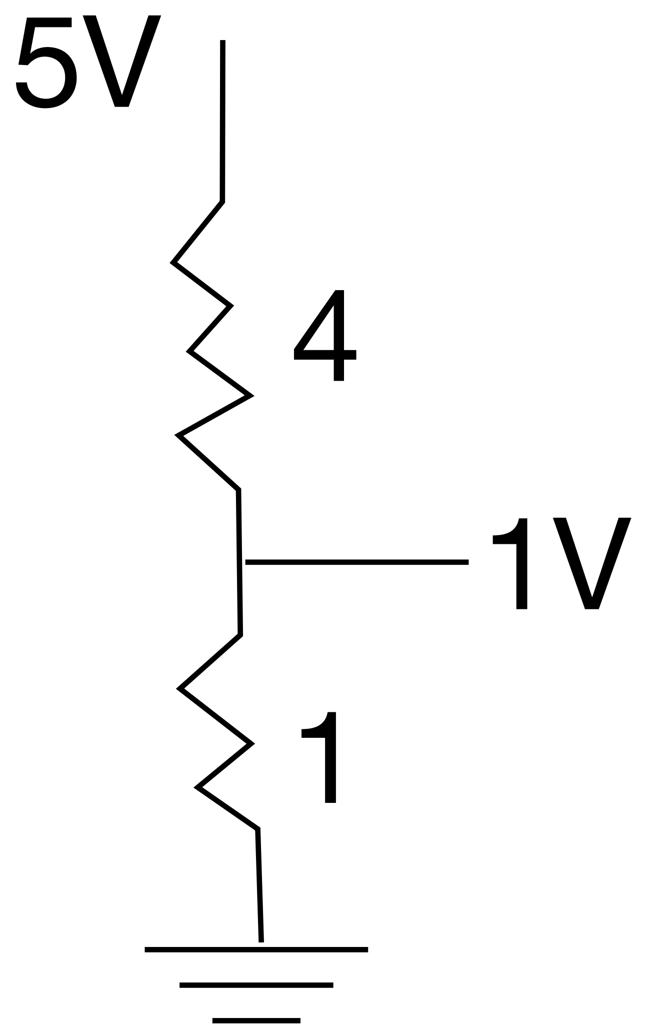
- In n-channel device, when gate-to-source voltage < threshold (2V), no current flows.
- Q1 acts as a load resistor: always in the conduction state.
- Q2 as the active device: encodes the logic of the circuit. Its behaviour will change with the voltage applied on A.
- When $A < V_T$, Q2 turns off. Since Q1 is always on, $Y > V_T$

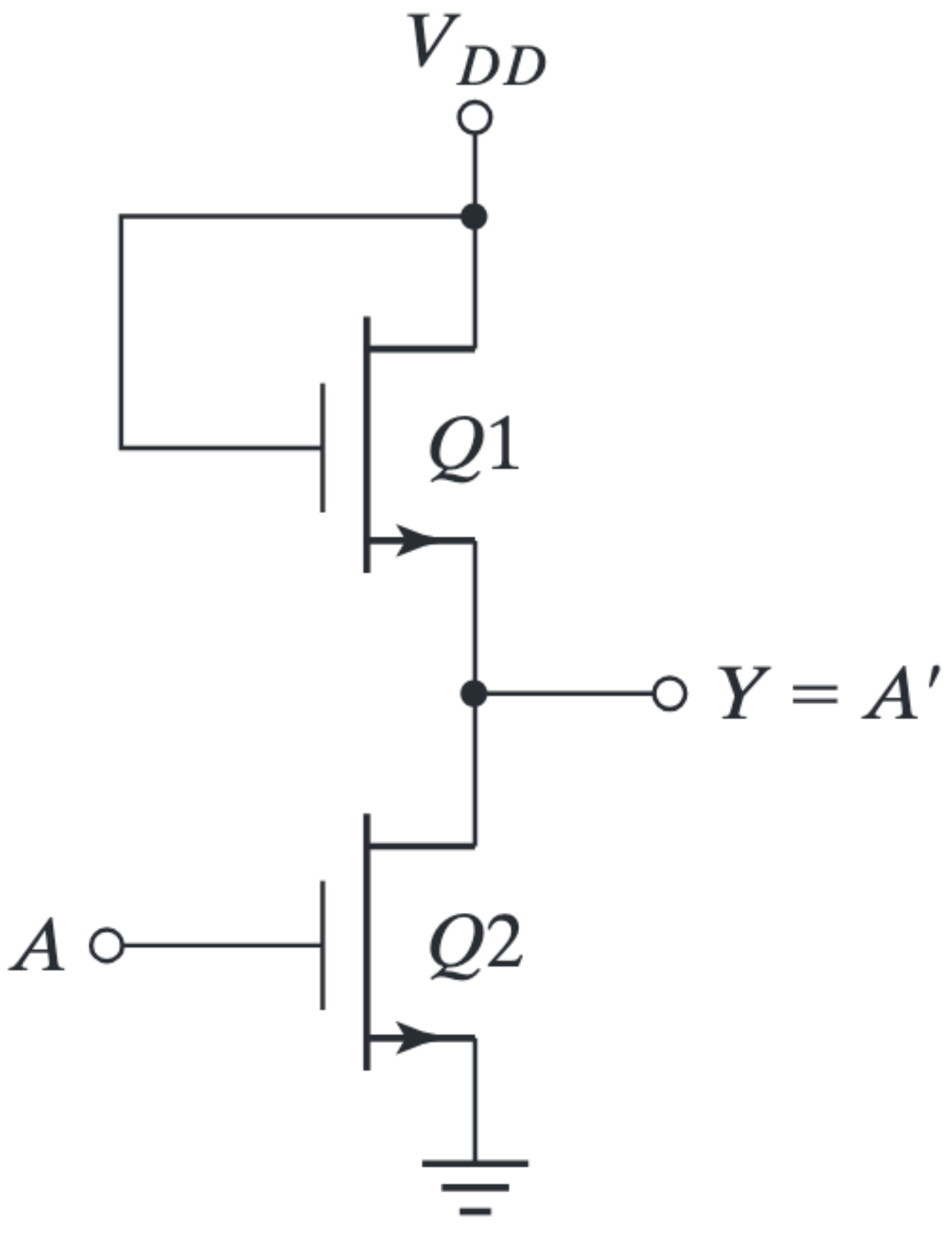


(a) Inverter

n-channel inverter (2)

- When $A > V_T$, Q2 turns on. Current flows from V_{DD} via Q1 and Q2. Resistance of Q2 is less than resistance of Q1 (1:4 in example below), to ensure that $Y < V_T$

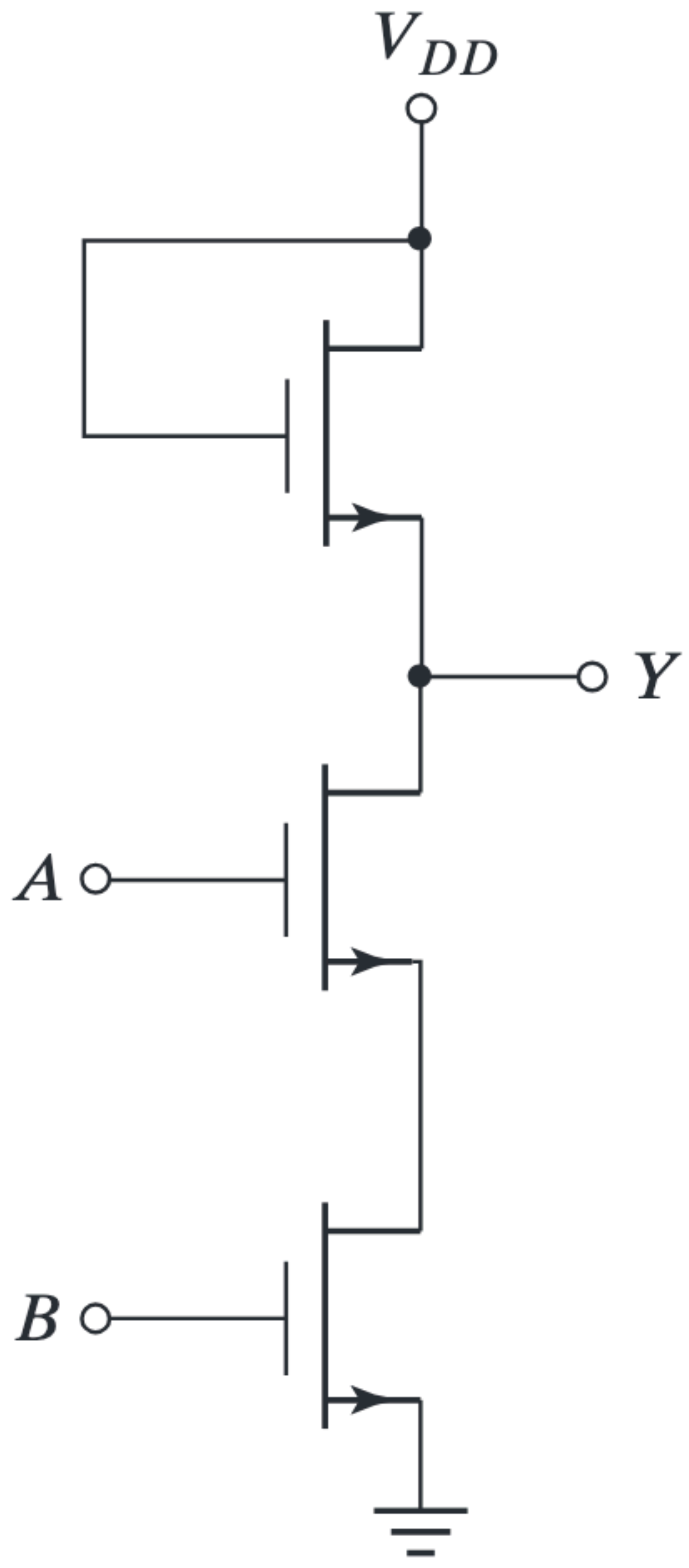




(a) Inverter

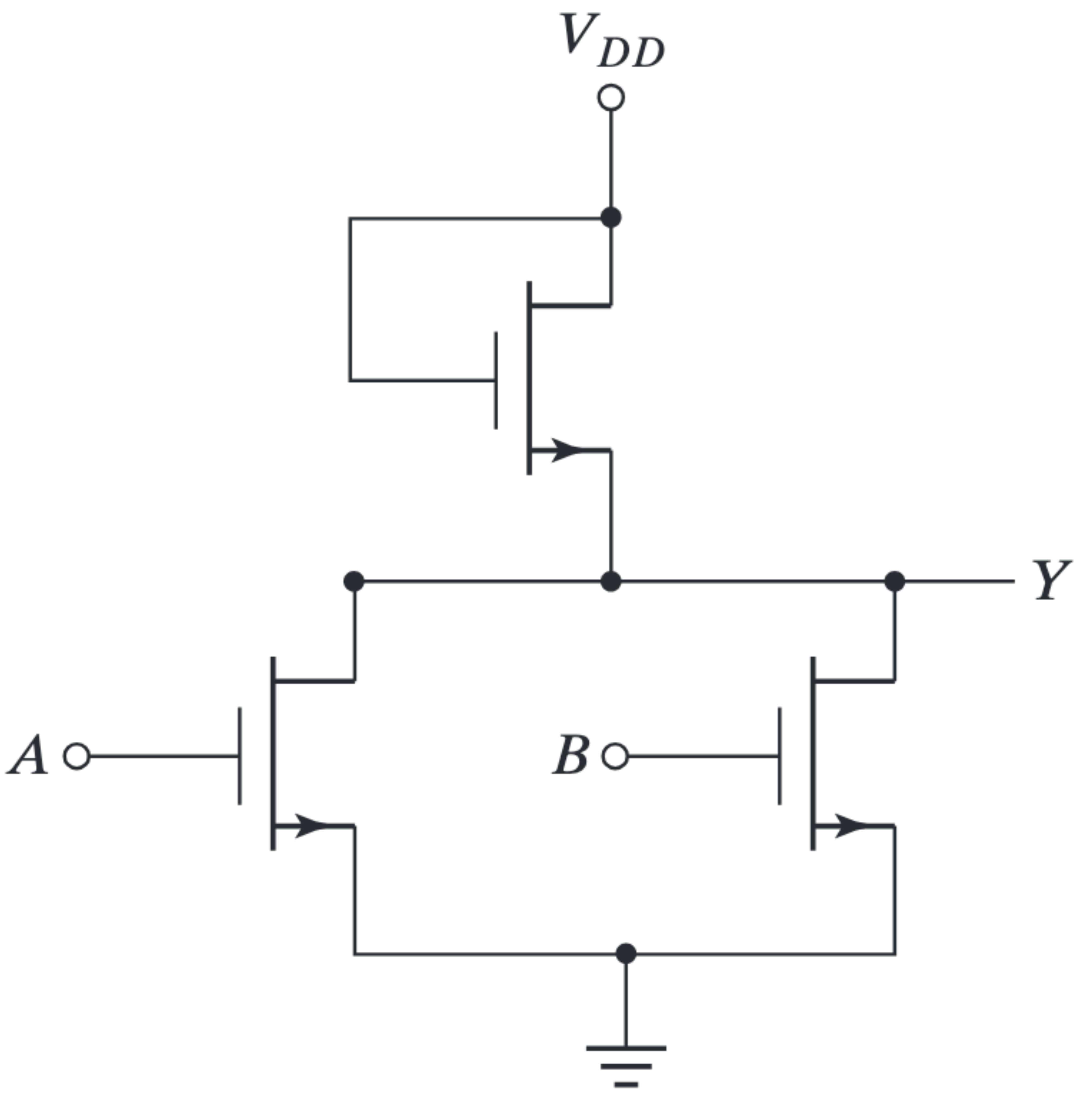
n-channel NAND gate

- When $A > V_T$ and $B > V_T$, $Y < V_T$
- When $A < V_T$ and $B > V_T$, Y ? V_T
- When $A > V_T$ and $B < V_T$, Y ? V_T
- When $A < V_T$ and $B < V_T$, Y ? V_T



n-channel NOR gate

- When $A > V_T$ and $B > V_T$, $Y < V_T$
- When $A < V_T$ and $B > V_T$, Y ? V_T
- When $A > V_T$ and $B < V_T$, Y ? V_T
- When $A < V_T$ and $B < V_T$, Y ? V_T



p-channel circuits

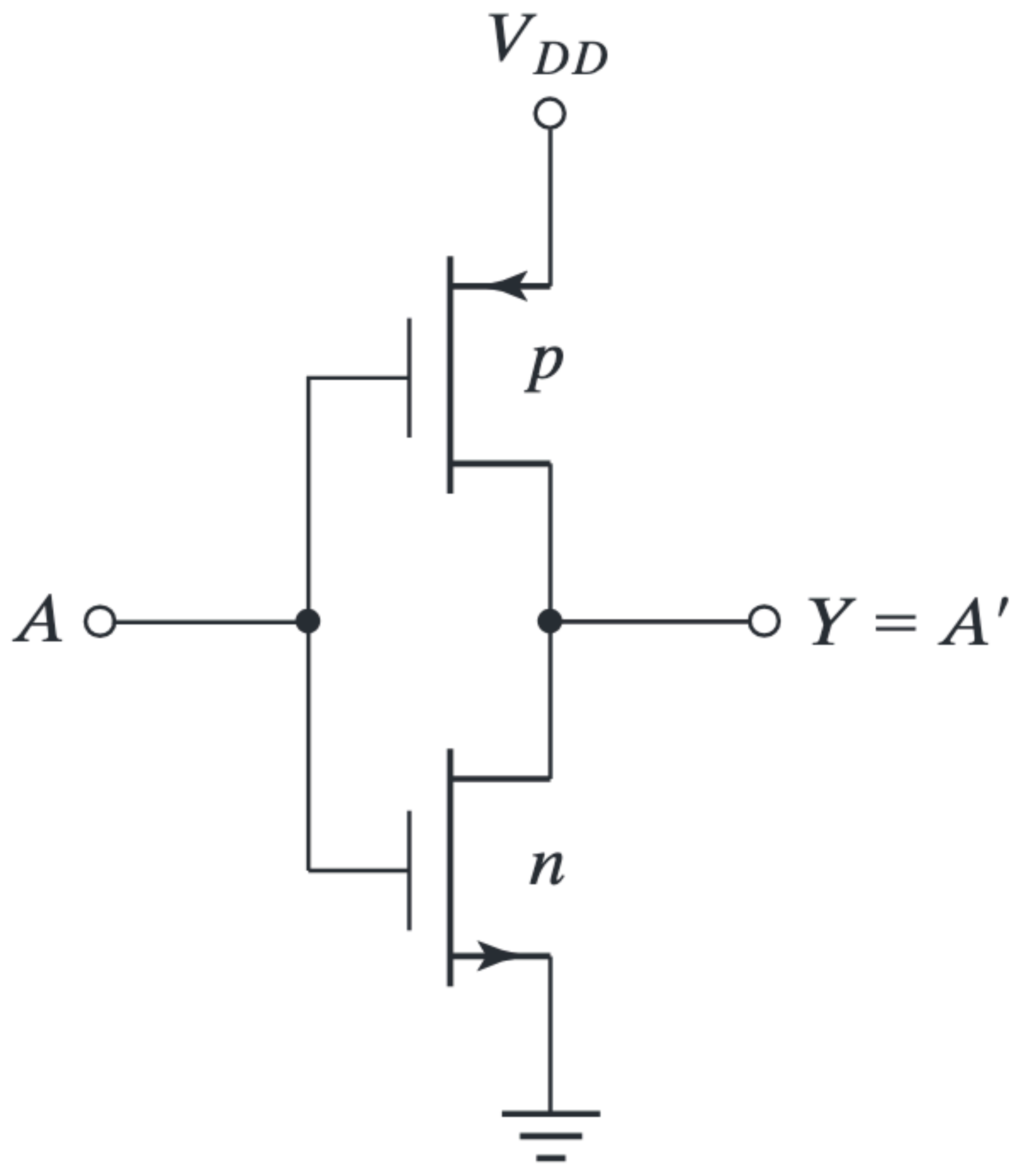
- In n-channel device, when gate-to-source voltage < threshold (2V), no current flows.
- In p-channel device, when gate-to-source voltage > threshold (-2V), no current flows.
- Inverter?
- NAND gate?
- NOR gate?

Complementary MOS (CMOS)

- Both n-channel and p-channel transistors can be fabricated on single silicon crystal
- CMOS circuits contain both types of transistors
- Source terminal of p-channel transistor is at V_{DD} (usually +5V).
  Source terminal of n-channel transistor is at GND

Inverter revisited

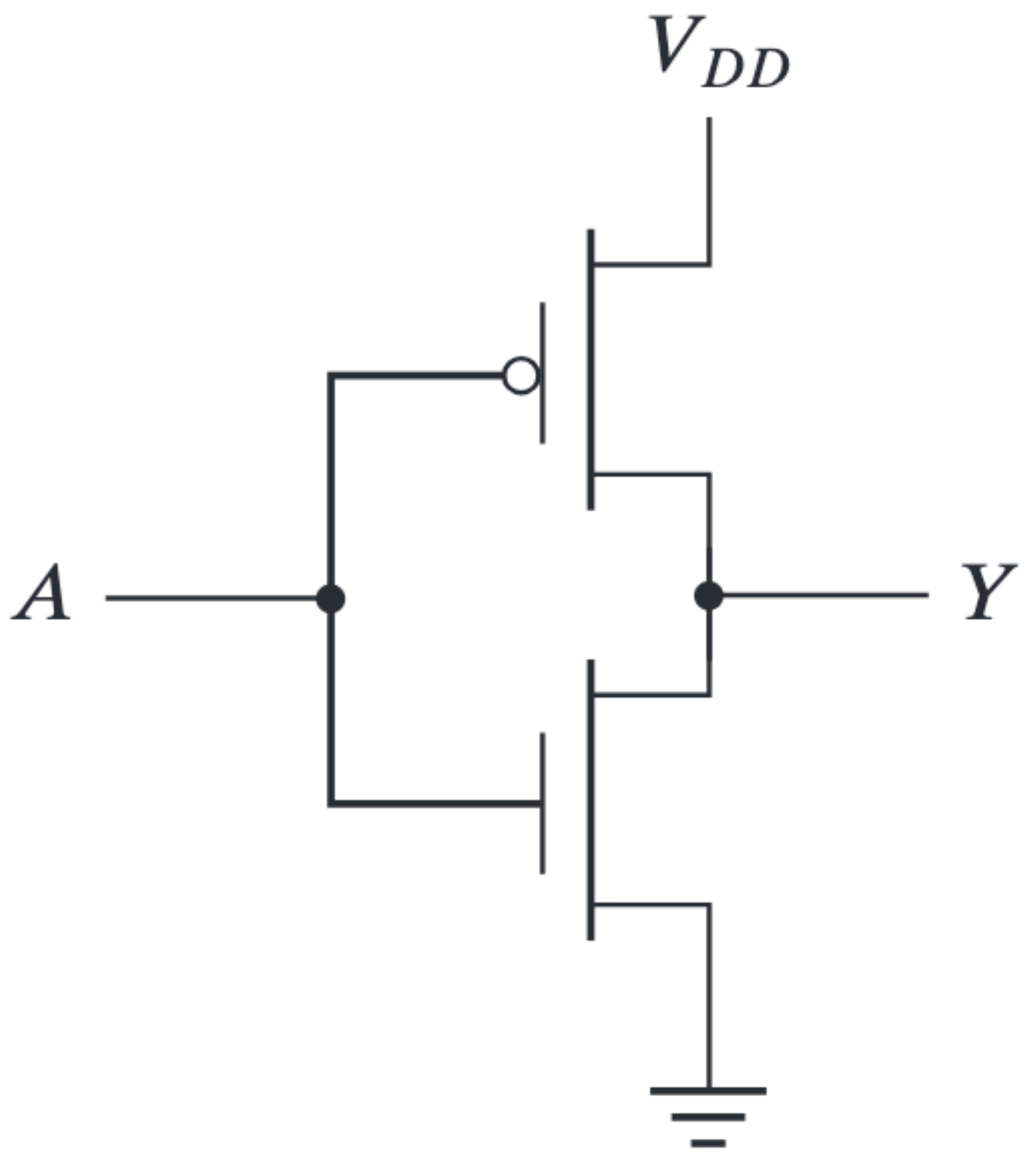
- n-channel MOS conducts when its gate-to-source voltage is positive.
- p-channel MOS conducts when its gate-to-source voltage is negative.
- Neither device conducts when its gate-to-source voltage is zero.
- When $A = V_{DD}$, Y = ?
- When A = GND, Y = ?



(a) Inverter

Logical model

- Instead of writing n and p, and drawing arrows on source, draw a bubble on gate for p-channel transistors
- Bubble represents that the gate conducts at low voltage
- When $A = V_{DD}$, Y = ?
- When A = GND, Y = ?



(b) Logical model

CMOS circuits

Most popular technology for digital circuits

- In a static state, power dissipation is very low (0.01mW). At least one transistor is always off in the path between power supply and ground when the state of the circuit is not changing.
- If the circuit is changing state, power dissipation increases. At 1MHz (1 mW), at 10MHz (5mW).
- CMOS fabrication process provides great packing density, i.e., more digital logic can be placed on a given area of silicon.